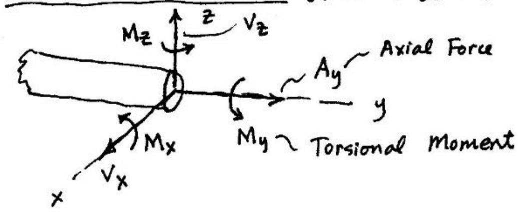
Statics: Intro to Internal Forces (Hibbeler Ch 7)

STATICS : INTERNAL FORCES (Hibbeler Chapter 7)

I. INTRO : Main goal of statics is to find forces everywhere in (on) a structure.
We have found: A. Resultant Force, Moment : of applied loads
B. Support Reactions: Cut structure free of supports to find these.
C. Internal forces (at pins) in multi-membered trusses and frames.
Key Step: Cut, or disassemble a pin connection to expose forces.
D. Now: Forces (and torques + moments) internal to a single member --
Key Step: Cut member at location of interest.

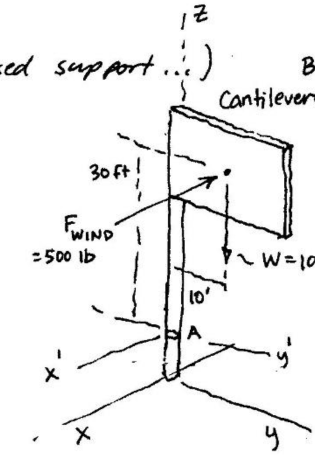
II. CASES

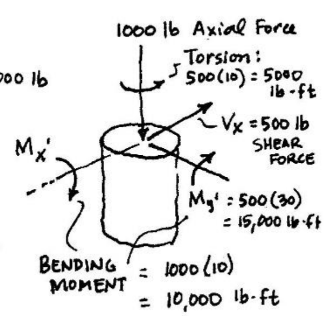
A. General 3-D Case (Recall a 3-D fixed support...)



Where, V_x, V_z are "Shear" forces
M_x, M_z are "Bending Moments"

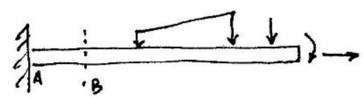
B. 3-D Example
Cantilevered Billboard.
Find internal forces at A:



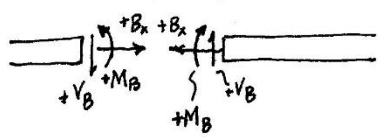


B. Axial Only } See handout.
C. Torsion Only }

D. Beams (General Info):



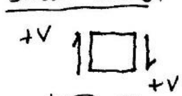
Cut At B:
Notice these are equal and opposite...



SIGN CONVENTION:

Axial: (Tension +)
(Compression −)

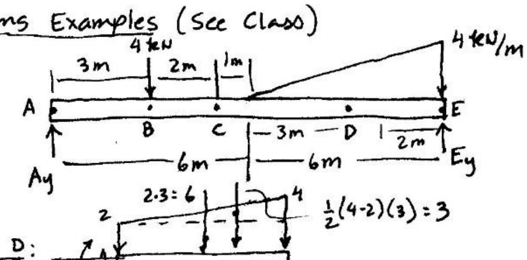
Shear Force: +V : Up on Left face, Down on right face,



Beam element has Clockwise Rotation

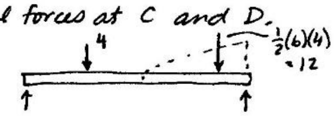
Bending Moment:

Beams Examples (See Class)



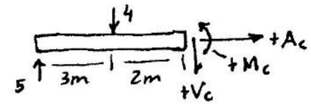
Find: Internal forces at C and D.

1. Support Rx's:



$A_y = \frac{3}{4}4 + \frac{1}{6}12$ $E_y = 11$ kN
$= 5$ kN

2. At C:



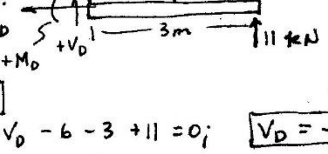
$\Sigma F_x = 0;$ $\boxed{A_c = 0}$

$+\uparrow \Sigma F_y = 0;$ $5 - 4 - V_c = 0;$ $\boxed{V_c = +1 \text{ kN}}$

$\circlearrowleft+ \Sigma M_{cut} = 0;$ $-5(5) + 4(2) + M_c = 0$

$\boxed{M_c = +17 \text{ kN}\cdot\text{m}}$

3. At D:



$\boxed{A_D = 0}$

$+\uparrow \Sigma F_y = 0;$ $V_D - 6 - 3 + 11 = 0;$ $\boxed{V_D = -2 \text{ kN}}$

$\circlearrowleft+ \Sigma M_{cut} = 0;$ $M_D + 6(1.5) + 3(2) - 11(3) = 0$

$\boxed{M_D = +18 \text{ kN}\cdot\text{m}}$